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EDITORIAL

Updates to the Minimum Information Required for A Glycomics Experiment (MIRAGE) guidelines

Carsten Kettner[1] · **Joseph Zaia**[2]

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Glycosylation diversifies the structures and functions of proteins. In eukaryotes, biosynthetic enzymes of the secretory pathway produce heterogeneous glycosylation built on core structures. Glycoprotein glycosylation serves as a protein folding quality control handle and as a binding site for lectin domain-containing proteins. Because such proteins, present in intracellular vesicles, on the plasma membrane, and in extracellular matrices, participate in all aspects of physiology, their analytical characterization remains a high priority, despite the challenges of their heterogeneous structures that require tailored analytical methods. Further, because the activities of the glycosylation biosynthetic enzymes that assemble glycans reflect numerous factors, including substrate concentrations and transport kinetics, it is necessary to analyze the distribution of glycan variants that exist on a given protein site. For this purpose, separation modes often interfaced with mass spectrometry are applied including liquid chromatography, capillary electrophoresis, and ion mobility spectrometry.

The Minimum Information Required for A Glycomics Experiment (MIRAGE) initiative aims to standardize reporting and dissemination of reproducible glycoanalytical methods. Hosted by the Beilstein-Institut, these guidelines follow the examples and share the fundamental criteria as applied in proteomics (MIAPE) [1], enzymology (STRENDA) [2], microarray (MIAME) [3, 4], metabolomics (MSI) [5–8], and lipidomics (LSI) [9]. The first MIRAGE criterion, required for a broad acceptance by the science community, is sufficiency. This guideline supports researchers to describe both the experimental setup and the data in a comprehensive way that enables individuals to understand, evaluate, interpret, and reproduce the data. The second criterion is practicability, meaning that the guidelines are concise and understandable, enabling researchers to report the data in compliance with minimal burden. The third and final criterion, stability, refers to long-term accessibility and maintenance of the guidelines with regards to the accommodation of technical and scientific advances. To date, the MIRAGE commission working groups have developed guidelines that reflect collective agreement from the glycoscience community that include sample preparation [10], mass spectrometry [11], glycan microarrays [12, 13], liquid chromatography (LC) [14] and NMR (https://doi.org/10.3762/mirage.5 and https://doi.org/10.3762/mirage.6). Most recently, members of MIRAGE published updated reporting guidelines for mass spectrometry experiments in glycoscience [15]. Readers can find additional information on these guidelines from the Beilstein Institute http://www.beilstein-mirage.org.

Analytical and Bioanalytical Chemistry published a recent topical collection "Tools and databases for studying data in the glycosciences" that summarizes advances in glycobioinformatics methods [16]. Figure 1 illustrates the centrality of MIRAGE reporting guidelines to glycomics and glycoproteomics analytical experiments. Data are submitted to a repository according to the FAIR (Findable, Accessible, Interoperable, and Re-usable) data principles [17]. Glycan structures are represented using glycan-specific notations including WURCS and GlycoCT or by chemical notations including IPUAC, SMILES, InChI and others [18]. The GlySpace Alliance [19] works collaboratively and synergistically with glycoscience portal projects to assure stable and accessible research tools without duplication of effort. Glyspace Allience members support WURCS and GlycoCT encoding while PubChem and ChEBI use InChI, SMILES and IUPAC as structural descriptors.

In glycomics, two data repositories have been developed, GlycoPOST [20] and UniCarb-DR [21], and the

✉ Carsten Kettner
ckettner@beilstein-institut.de

✉ Joseph Zaia
jzaia@bu.edu

[1] Beilstein-Institut, Frankfurt Am Main, Germany

[2] Department of Biochemistry & Cell Biology, Chobanian & Avedisian School of Medicine, Boston University, Boston, MA, USA



Fig. 1 The MIRAGE guidelines are central to public tools for processing glycoanalytical data

data submission cross-referencing systems for both have been unified [22] and are available through the GlyCosmos Portal [23]. The GlyTouCan [24, 25] repository assigns accession numbers to glycan structures and the GlyComb repository [26] to glycoconjugates. These tools provide consistency of glycoanalytical data reporting and use. GlycoStore [27] is a curated database of chromatographic, electrophoretic and mass spectrometry composition information built from public data sets that supports glycoanalytical projects. The Carbohydrate microarray Analysis and Reporting Tool (CarbArrayArt) [28] is a repository for glycan microarray data management. The GRITS Toolbox [29] archives, processes and interprets mass spectrometry-based data on released glycans.

Analytical scientists who use glycoanalytical methods benefit from using the MIRAGE guidelines that define the minimum information required for reporting and facilitate sharing and re-use and effective mining of the data central to glycobioinformatics. In addition, scientists that use MIRAGE compliant publications for their further work will benefit from a standardized dataset that provide all data and metadata required for reuse and reproduction. These reporting guidelines are supported by *Analytical and Bioanalytical Chemistry*.

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Carsten Kettner studied biology at the University of Bonn and obtained his diploma from the University of Göttingen. In 1999, he was awarded his PhD for his work on the biophysical comprehension of the yeast vacuolar ATPase through applying electrophysiological techniques at the University of Karlsruhe. As a post-doctoral student, he continued both his studies on the biophysical properties of the pump and his investigations of the kinetics and regulation of the plasma membrane potassium channel. In 2000, he joined the Beilstein-Institut. Here, he is responsible (a) for the organization of the Beilstein symposia, (b) for the project management of the data standardization initiatives, and (c) for the marketing of Beilstein-Institut projects, especially the Beilstein journals.

In 2007, he was awarded his certificate of competence as a project manager for his studies and thesis by the Studiengemeinschaft Darmstadt (a certified service provider). Since 2004 he has coordinated the work of the STRENDA commission and promoted guidelines for reporting enzyme data along with the commissioners (www.strenda.org). Today, these reporting standards have been adopted by over 55 biochemical journals for their instructions for authors and are incorporated into the electronic data validation and storage tool, STRENDA DB.

Since 2011, Carsten has co-ordinated the MIRAGE project, which aims to establish uniform reporting and representation of glycomics data in publications and which became an essential hub for the development of glycomics infrastructure (www.beilstein-mirage.org). In 2014, Carsten was appointed head of the funding and conferences department, which is also in charge of the foundation's public relationships. He is very interested in any aspects of open science and open access, and thus, he is actively engaged in a number of initiatives that are dedicated to making scientific data open and accessible to the wider public, including NFDI4Chem, Force11 and RDA (Chemical Data, FAIRsharing).



Joseph Zaia serves as Professor in the Department of Biochemistry & Cell Biology at the Boston University Chobanian & Avedisian School of Medicine. His primary role is as a research mentor for a group of undergraduate and graduate students and postdoctoral scientists. His research concerns the biochemistry of extracellular matrix molecules, including glycoproteins and proteoglycans, using mass spectrometry-based proteomics, glycomics, glycoproteomics and bioinformatics. Dr. Zaia has published more than 190 articles in peer-reviewed scientific journals. He received the 2020 Journal of the American Society for Mass Spectrometry reviewer award and the 2021 American Society for Biochemistry and Molecular Biology Molecular and Cellular Proteomics lectureship award.

Before joining Boston University in 1999, he worked at Osiris Therapeutics, on the molecular basis of connective tissue diseases. His research group focusses on glycomics and glycoproteomics of viruses and neuropsychiatric and neurodegenerative diseases. His group developed mass spectrometry-based methods for the characterization of glycosaminoglycans and applied these to collaborative biomedical studies. Using these methods, they routinely analyze four glycan classes plus proteins from a tissue slide. His group also developed a bioinformatics program, known as GlycReSoft, for the interpretation of glycomics and glycoproteomics liquid chromatography-mass spectrometry data sets. GlycReSoft uses averagine values tailored to glycopeptides to pick peaks from the raw mass spectral dataset. It combines proteomics and glycomics data to generate a search space against which the picked peaks are searched. GlycReSoft allows users to identify with high confidence site-specific glycosylation of complex glycoproteins.

In addition to his role as Editor of *Analytical and Bioanalytical Chemistry*, Joseph Zaia serves on the editorial boards of *Molecular and Cellular Proteomics*, *Journal of the American Society for Mass Spectrometry*, and *Mass Spectrometry Reviews.* He has taught tutorials on glycomics and glycoproteomics at conferences including US HUPO, the ASMS Conference on Mass Spectrometry and Allied Topics, and the Society for Glycobiology Annual Conference.